

YUVARAJ S

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LinkedIn

GitHub

Portfolio

Professional Summary

AI/ML Engineer with specialized expertise in computer vision, natural language processing, and full-stack integration. Demonstrated success in developing production-ready machine learning solutions that solve real business challenges. Skilled in optimizing AI models for real-time applications with a focus on practical implementation and measurable performance metrics.

Skills

Technical

- TensorFlow, PyTorch, Scikit-learn
- Knowledge on Git & GitHub
- C, Python
- HTML, CSS
- UI Designing - Figma
- Machine Learning Algorithms
- OpenCV, YOLO
- BERT, NLP, Transformers
- Django, REST APIs
- AWS AI Foundations

Non-Technical

- Active Learner
- Innovation and Creativity
- Good Communication
- Leadership
- Problem Solving
- Logical Reasoning
- Team Collaboration
- Time Management
- Adaptability

Professional Experience

AI Machine Learning Engineer

July 2024 – October 2024

Techso IT LLC USA

- Engineered a Django-based real-time audio transcription system using SpeechRecognition , achieving 83%+ transcription accuracy on enterprise audio datasets
- Developed and deployed NLP-driven text summarization and multilingual translation modules (BART & mBART), reducing content review time by 40% and enabling seamless cross-language communication

Hackathons & Competitions

AI-Web Development Engineer

October 2024

Flipkart Robotic Challenge National Competition

- Designed and integrated 5 different AI models into unified platform with <100ms latency
- Developed RESTful API endpoints for seamless AI-frontend communication

Machine Learning Developer

September 2024

Amazon ML Challenge National Competition

- Developed NLP models to extract product attributes from unstructured text with 82% accuracy
- Implemented BERT-based feature extraction techniques for e-commerce product categorization

AI Solutions Developer

July 2023

IBM Hackathon Regional Competition

- Created facial emotion recognition system with real-time processing capabilities
- Built recommendation algorithms connecting detected emotions with personalized suggestions

Projects

Automated Library Access & Attendance System <i>Android & Django</i>	Nov 2024 – Apr 2025
• Built a mobile check-in/checkout system using ID scan and face capture for secure access. • Created a central backend to record entries/exits and manage staff/students and departments. • Cut manual logging time by 85% and gave teams real-time visibility into library usage. • Added easy CSV reports (daily, monthly, yearly, custom) and simple bulk data upload.	
FlipVision AI: E-Commerce Personalization <i>Computer Vision & ML</i>	October 2024 – Present
• Designed and integrated 5 different AI models into unified platform with <100ms latency • Developed RESTful API endpoints for seamless AI-frontend communication	
AI-Powered Appointment Booking Chatbot <i>Django & Conversational Flow</i>	August 2024 – September 2024
• Developed a multi-step chatbot to guide patients through booking medical appointments seamlessly. • Integrated department and doctor selection, real-time slot availability checks, and symptom collection. • Implemented appointment scheduling with slot validation, automatic booking, and confirmation handling. • Improved scheduling efficiency and reduced manual staff workload by automating the entire booking flow.	
EchoFlow: Audio Analysis Platform <i>NLP & Signal Processing</i>	July 2024 – August 2024
• Designed multi-stage audio processing pipeline for real-time transcription and analysis • Implemented BERT-based sentiment analysis on transcribed content with 87% accuracy	
Mentor Selection System <i>Django & Web Development</i>	February 2023 – October 2023
• Developed comprehensive web application for managing mentor-learner relationships with role-based access control • Implemented Django backend with HTML, CSS, Bootstrap frontend supporting learner-mentor matching and HOD oversight	

Education

B.Tech in Artificial Intelligence and Machine Learning	August 2021 – May 2026
<i>Saveetha Engineering College</i>	<i>Chennai, India</i>
• Specialization: Deep Learning Architectures, Computer Vision, Natural Language Processing • Research Focus: Multi-modal Learning and Real-time AI Systems for Practical Applications • Key Coursework: Advanced Machine Learning, Neural Networks, Computer Vision, NLP, Data Structures • Active member of AI/ML community with 200+ members, participating in workshops and coding competitions	

Professional Certifications & Achievements

Efficient Large Language Model (LLM) Customization — NVIDIA — May 2025
AWS Certified AI Practitioner — Amazon Web Services — February 2025
Oracle Cloud Infrastructure 2024 AI Foundations Associate — Oracle — January 2025
Deep Learning Specialization — Coursera — November 2023 — Including: CNNs, TensorFlow, NLP
Secured top rankings in national-level AI/ML competitions and hackathons